

Reviewer Pack — Minimal Quadratic Residue Representation and Communication C...

Entry 4ca114e86209 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Quadratic Residue Representation and Communication Complexity Correlation

Entry ID: 4ca114e86209 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-31 23:43:21 UTC

1. Conjecture

Field A (mathematical branch): Quadratic Residue Theory **Field B** (complexity object): Communication Complexity

Statement:

The minimal quadratic residue representation of a communication complexity instance φ is linearly correlated with its communication complexity, such that the number of unique quadratic residues used in the minimal representation is $\Theta(\text{CommunicationComplexity}(\varphi))$.

Rationale (proposer's reasoning):

Quadratic Residue Theory has not been extensively applied to Communication Complexity. This conjecture aims to expose potential structural connections between these areas by proposing a quantitative relationship. If true, it could provide new insights into communication complexity and potentially lead to new algorithms or complexity lower bounds.

Taxonomy category: new_conjecture (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 4528c3bd75f507ba

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The number of unique quadratic residues in the minimal representation is within a factor of 1.5 from the $\text{CommunicationComplexity}(\varphi)$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	1.00	SAFE	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=0, elapsed=0.0s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def communication_complexity(n):
        # Placeholder for actual communication complexity calculation
        return n # Simplified example

    def minimal_quadratic_residue_representation(n):
        residues = set()
        for i in range(1, n + 1):
            residues.add(i * i % n)
        return len(residues)

    n_values = [5, 10, 15, 20, 30, 40]
    results = []

    for n in n_values:
        cc = communication_complexity(n)
        mqr = minimal_quadratic_residue_representation(n)
        results.append({"n": n, "cc": cc, "mqr": mqr})

    if not results:
        return {
            "metric_name": "Correlation",
            "metric_value": None,
            "instances_tested": 0,
            "n_max": 0,
            "conjecture_holds": False,
            "counterexample": "No instances generated"
        }

    mean_cc = sum(result["cc"] for result in results) / len(results)
    mean_mqr = sum(result["mqr"] for result in results) / len(results)
    correlation = (mean_cc * mean_mqr - mean_cc * mean_mqr) / (math.sqrt(mean_cc ** 2 * mean_mqr ** 2))
```


8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The test only considers n values up to 40, which is too small to draw a meaningful conclusion about the conjecture’s validity. The metric does not scale trivially with n, but the sample size is insufficient for reliable empirical testing.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The test results indicate that the conjecture does not hold for all instances tested within the given range (n_max = 40). The correlation metric value | next: Further investigation is needed to determine if the conjecture holds for larger values of n or under different conditions.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14068
2	preregistration	ollama_remote	glm4:latest	0	0	9654
3	novelty	ollama_remote	glm4:latest	0	0	8431
4	novelty	ollama_remote	glm4:latest	0	0	8268
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13785
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	17712
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8146
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8392
9	critic	ollama_remote	glm4:latest	0	0	16504
10	judge	ollama_remote	glm4:latest	0	0	11993

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 116953 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in research/audit/4ca114e86209.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/4ca114e86209.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/4ca114e86209.tar.gz (if generated)